

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Applied AI Engineer

Applied AI engineer with 3+ years building production ML systems in Python, now focused on RAG, LLM evaluation, and context engineering. Built and evaluated a hybrid retrieval pipeline (BM25, dense, RRF, cross-encoder) for financial-document QA, benchmarked against Snowflake Cortex Search and exposed through typed MCP tools for programmatic failure diagnosis. Strong on evaluation and reliability: Pydantic-validated LLM-as-judge contracts and reproducible experiment runs.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built a **hybrid-retrieval** RAG pipeline for financial-document QA (**BM25, dense embeddings, reciprocal rank fusion, cross-encoder reranking**), with three isolable **retrieval** modes (closed-book, oracle-context, dense) to separate retrieval quality from generation quality.
- Designed a dual-track evaluation layer pairing deterministic checks (exact/normalised match, scale-aware numeric matching e.g. "\$1.5B" vs "1,500 million") with an **LLM-as-judge**, and a retrieval-vs-answer metric join that attributes each wrong answer to a retrieval miss, generation error, or grounding failure.
- Benchmarked the custom pipeline against **Snowflake Cortex Search** on identical chunks, embeddings, and metrics, and used **Pydantic** v2 schema validation across provider configs, judge outputs, and **MCP/FastAPI** request-response models for robust, typed API boundaries.
- Exposed runs, retrieval results, metrics, and failure inspection through typed **MCP** tools and resources with Markdown/CSV/JSON reporting and a compare-runs mode, enabling AI clients to diagnose failures programmatically.

WORK EXPERIENCE

Independent Projects & Upskilling

Sep 2024 - Present

Self-directed

- Built and evaluated a hybrid-retrieval RAG pipeline for financial-document QA (BM25, dense retrieval, RRF, cross-encoder reranking, LLM-as-judge scoring), benchmarked against Snowflake Cortex Search and exposed through typed MCP tools for programmatic diagnosis. Full detail in Projects.
- Completed Nvidia "Building RAG Agents with LLMs" (Jul 2026), covering RAG, agents, and evaluation with LangChain and LangSmith.

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Presented model performance, failure modes, and deployment trade-offs to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries and John Deere.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Worked with hardware, software, QA, and client teams to translate model performance, latency, and failure analysis into rollout decisions for deployed safety systems.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Jul 2026

Nvidia

Deep Learning Specialisation

Jul 2019

deeplearning.ai

SKILLS

- **AI-Assisted Development:** Claude Code, ChatGPT Codex
- **AI Engineering and Evaluation:** LLM-as-judge, RAG evaluation, document-grounded QA, retrieval metrics, evidence matching, citation checks, hallucination detection, Prompt engineering, error analysis, structured outputs, RAG pipelines, dense retrieval, FAISS vector store, ChromaDB, hybrid retrieval, Pydantic v2, FastAPI, MCP
- **Experimentation and Reliability:** failure-mode analysis, benchmarking, regression checks, hypothesis testing, ablation studies, post-deployment validation, production monitoring
- **Python and ML Engineering:** Python, Pandas, NumPy, scikit-learn, PyTorch, MLflow, DVC, Apache Airflow, Optuna, LangChain, LangSmith, PyDantic, FastAPI, Snowflake
- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux, schema-validated output contracts, Microsoft Azure, SQL